

SNMP and the Management Information Base (MIB)

1 The Simple Network Management Protocol (SNMP)

The **SNMP protocol** is an application-layer protocol used to exchange management information between a managing server and an agent.

Operational Modes

- **Request-Response Mode:** The manager sends a request (Get/Set) to the agent, who performs the action and replies.
- **Trap Mode (Unsolicited):** The agent sends a **trap message** to notify the manager of an exceptional event (e.g., link interface going up or down).

1.1 SNMPv3 PDU Types

SNMPv3 defines seven types of **Protocol Data Units (PDUs)**:

PDU Type	Sender	Description
GetRequest	M → A	Retrieve one or more MIB object instances.
GetNextRequest	M → A	Retrieve the next MIB object instance in a list/table.
GetBulkRequest	M → A	Retrieve a large block of data (e.g., a large table).
SetRequest	M → A	Set the value of one or more MIB object instances.
InformRequest	M → M	Inform a remote managing entity of MIB values.
Response	A → M	Generated in response to a Get/Set/Inform request.
SNMPv2-Trap	A → M	Inform the manager of an exceptional event.

Table 1: SNMPv3 PDU Types (M=Manager, A=Agent).

Transport and Reliability

- **Preferred Transport: UDP (RFC 3417).**

- **Request ID:** Used to match responses to requests and detect lost packets.
- **Security:** SNMPv3 added critical security and administration features, shifting the focus from simple monitoring to active control (like **SetRequest**).

2 The Management Information Base (MIB)

Managed objects are gathered into **MIBs**. These objects can be counters (IP datagrams discarded), descriptive info (software version), or status info.

SMI (Structure of Management Information)

SMI is the data description language used to define MIB objects. It ensures that the syntax and semantics of management data are unambiguous.

2.1 Example: MIB Object Definition

The following is an example of an **OBJECT-TYPE** definition from RFC 4293:

ipSystemStatsInDelivers

```
ipSystemStatsInDelivers OBJECT-TYPE
    SYNTAX          Counter32
    MAX-ACCESS      read-only
    STATUS          current
    DESCRIPTION
        "The total number of datagrams successfully
        delivered to IP user-protocols (including ICMP)."
```

::= { ipSystemStatsEntry 18 }

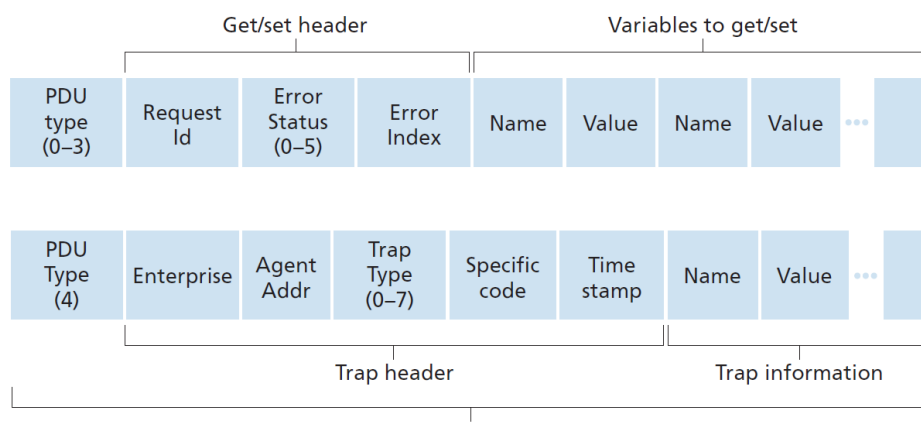


Figure 1: Structure of the SNMP PDU: Get/Set header vs. Trap header.